SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 30	Semester: January 2025-April 2025 Examination: In-Semester Examination	Duration :1 hr15min
Programme code: 01 Programme: B Tech	Class: SY	Semester: IV (SVU 2023)
Institute/School/ Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 216U01C404	Name of the Course: Operating Systems	

Question No.		Max. Marks	CO	BT
Q1	Solve Any ***THREE***	**15**		

A	With respect to Process creation, examine the program given below and predict the output for the program with proper justification.
	#include <stdio.h>
	#include <sys/types.h>
	#include <unistd.h>
	void forkexample()
	{
	int x=13,y =37;
	if (fork() == 0)
	printf("Child has x = %d and y=%d\n", ++x,--y);
	else
	printf("Parent has x = %d and y=%d\n", --x,++y);
	}
	
	int main()
	{
	forkexample();
	return 0;
	}
		05	CO1	UN, EV

B	List the different types of Preprocessor Directives. Explain Conditional Compilation Directives with the help of an example.	05	CO1	RE, UN
C	Enlist the operating system functions. Explain any two functions in brief.	05	CO1	RE
D	Discuss the process of System Boot with the help of a suitable diagram.	05	CO1	RE

Q2	Solve the ***ANY ONE***	**10**		
Q2A	i) Discuss the different multithreading models, including their advantages and disadvantages.	05	CO2	RE
	ii) Draw and explain the 5 state process models elaborating all possible state transitions.	05	CO2	RE, UN
	OR			
Q2B	Table below shows the set of processes with their arrival and burst time.	10	CO2	UN, AP

Process	Arrival Time	Burst Time
P1	0	3
P2	2	3
P3	2	2
P4	4	1

Calculate average waiting time and average turnaround time if the processes are scheduled using the following scheduling algorithms:

i) Shortest Job First Non-preemptive (i.e shortest process next)

ii)Multilevel Feedback Queue (with number of ready queues in the system =3 and time quantum for the queue are 1ms, 2ms, 4ms for Queues Q1,Q2,Q3 respectively).

Q3	Consider four memory partitions of size 200KB, 30KB, 750KB, and 50KB. These partitions need to be allocated to the processes given below:-	05	CO5	RE, AP

Process	Memory Required
P1	30KB
P2	200KB
P3	500KB
P4	45KB

Consider the Order for allocation to be taken as P1, P2, P3, P4 only.

How memory allocation would be performed using the **First Fit, Best Fit, and Worst Fit allocation methods.**